



Experiment 8.1

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1. Aim: Create Login Form with React State Management

2. Objective: Learn how to build a login form in React using basic state management with the useState hook. This task helps you understand how to capture and handle form input data, manage local component state, and respond to user actions in a React application.

4. Implementation/Code:

```
import React, { useState } from 'react'

export default function App() {
  const [username, setUsername] = useState("")
  const [password, setPassword] = useState("")
  const [error, setError] = useState("")

  const handleSubmit = e => {
    e.preventDefault()

    if (!username || !password) {
      setError('Please fill in both fields.')
      return
    }
  }
}
```

```
}
```

```
console.log('Username:', username)
console.log('Password:', password)
setError("")
alert('Login successful! (Check console for details)')
}
```

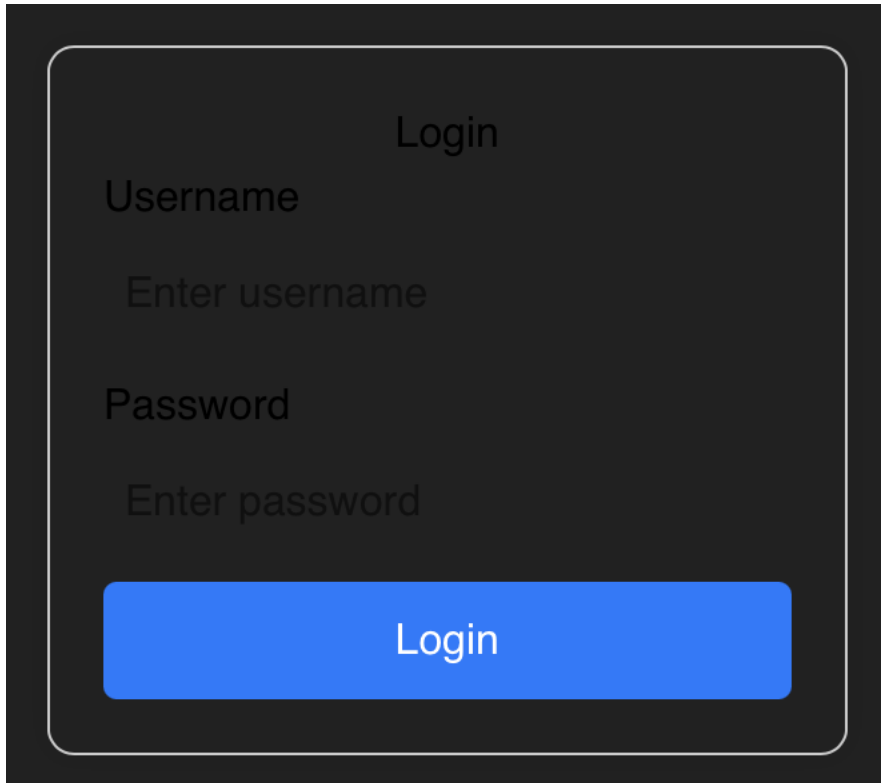
```
return (
  <div style={{
    display: 'flex', justifyContent: 'center', alignItems: 'center',
    height: '100vh', fontFamily: 'sans-serif'
  }}>
    <form
      onSubmit={handleSubmit}
      style={{
        width: 300, padding: 20, border: '1px solid #ccc',
        borderRadius: 10, boxShadow: '0 0 10px rgba(0,0,0,0.1)'
      }}
    >
      <h2 style={{ textAlign: 'center' }}>Login</h2>
      <div style={{ marginBottom: 10 }}>
        <label>Username</label><br />
        <input
          type="text"
          value={username}
          onChange={e => setUsername(e.target.value)}
          placeholder="Enter username"
        />
      </div>
    </form>
  </div>
)
```

```

        style={{ width: '100%', padding: 8, marginTop: 4 }}
    />
</div>
<div style={{ marginBottom: 10 }}>
    <label>Password</label><br />
    <input
        type="password"
        value={password}
        onChange={e => setPassword(e.target.value)}
        placeholder="Enter password"
        style={{ width: '100%', padding: 8, marginTop: 4 }}
    />
</div>
{error && <p style={{ color: 'red', fontSize: 14 }}>{error}</p>}
<button
    type="submit"
    style={{
        width: '100%', padding: 10, backgroundColor: '#007BFF',
        color: 'white', border: 'none', borderRadius: 5, cursor: 'pointer'
    }}
>
    Login
</button>
</form>
</div>
)
}

```

5. Output:



The image shows a dark-themed login form. At the top, the word "Login" is centered. Below it, there are two input fields. The first is labeled "Username" and has a placeholder text "Enter username". The second is labeled "Password" and has a placeholder text "Enter password". At the bottom of the form, there is a prominent blue button with the text "Login" in white.

Learning Outcome :

You will learn how to use React's useState hook to manage and validate form data, handle user input in real time, and respond to form submission events effectively within a component.